QUASI-REGULAR IMMERSIONS OF SPACES CHARACTERIZED BY FINITE PRESENTATION, FINITE LOCALLY FREE CONORMAL, AND THE CONORMAL-ALGEBRA ISOMORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 10604-10614 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 10616-10622, the complete author proof (5 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. One of the three flavour-dictionary cards cited by this wave's sibling L16157 (regular-immersion-noetherian) and last wave's conormal-sequence card L16129 — the flavour web on the spaces side completes with this and L16173-adjacent material.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter definition section for immersion flavours; chapter lemma etale-conormal-algebra (etale descent of the conormal algebra); Divisors, lemma quasi-regular-immersion (the schemes characterization).

Source reference. Original lemma 06BN: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 10530–10546. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 063M: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 10622–10622. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 09RQ: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 673–675. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 09RR: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 680–689. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let $i : Z \to X$ be an immersion of algebraic spaces over S. Then i is a quasi-regular immersion if and only if the following conditions are satisfied*

(1) *i is locally of finite presentation,*
(2) *the conormal sheaf $\mathcal{C}_{Z/X}$ is finite locally free, and*
(3) *the map (09RQ) is an isomorphism.*

Proof. Follows from the case of schemes (Divisors, Lemma 063M) via étale localization (use Definition 06BN and Lemma 09RR). □